AI HOME DESIGNER
DATABASE SPECIFICATION
PART 1

STATUS:
FOUNDATION DATABASE ARCHITECTURE

==
=========
1. DATABASE GOALS
==
=========

PRIMARY GOAL

A database shall:

- store every project
- store every room
- store every user
- store AI history
- store project versions
- store exports
- store permissions
- store audit logs

NON-FUNCTIONAL GOALS

✓ ACID compliant

✓ scalable

✓ easy backup

✓ high integrity

✓ optimized for read + write

==
=========
2. DATABASE ENGINE
==
=========

PRIMARY DATABASE

PostgreSQL 17+

Reason:

✓ mature

✓ JSONB support

✓ GIS support (future)

✓ excellent indexing

✓ replication

✓ reliability

CACHE

Redis

Purpose:

- session cache
- AI cache
- temporary storage

OBJECT STORAGE

S3 compatible

Stores:

- PDF exports
- images
- thumbnails
- backups

===

3. DATABASE DESIGN PRINCIPLES

===

RULE 1

Never duplicate business data.

RULE 2

Every table has UUID primary key.

RULE 3

Every table has:

created_at

updated_at

RULE 4

Soft delete where business relevant.

deleted_at

RULE 5

Never store calculated values unless caching.

==
=========
4. DATABASE NAMING CONVENTIONS
==
=========

TABLES

snake_case

Examples

users

projects

project_rooms

exports

COLUMNS

snake_case

Examples

created_at

owner_id

project_name

==
=========
5. PRIMARY ENTITY MAP
==
=========

CORE TABLES

users

organizations

projects

project_versions

rooms

walls

doors

windows

roofs

floors

materials

exports

audit_logs

ai_requests

ai_responses

permissions

sessions

notifications

==
6. ENTITY RELATIONSHIPS
==

ONE USER

↓

Many Projects

ONE PROJECT

↓

Many Versions

ONE VERSION

↓

Many Rooms

ONE ROOM

↓

Many Walls

ONE WALL

↓

Many Windows

ONE WALL

↓

Many Doors

===
=========
7. PRIMARY KEY STANDARD
===
=========

ALL TABLES

id UUID PRIMARY KEY

Example

id

550e8400-e29b-41d4-a716-446655440000

===
=========
8. COMMON COLUMNS STANDARD
===
=========

Every business table contains

id

created_at

updated_at

created_by

updated_by

deleted_at

Example

projects

id

name

created_at

updated_at

created_by

updated_by

deleted_at

===
=========
9. TIMESTAMP STANDARD
===
=========

Always

TIMESTAMP WITH TIME ZONE

UTC only

Frontend converts timezone.

===
=========
10. DATABASE FOUNDATION COMPLETE

===
=========

NEXT SECTION

Users

Organizations

Authentication

Permissions

===
=========
END OF DATABASE SPECIFICATION PART 1
===
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 2

STATUS:
IDENTITY, ORGANIZATIONS AND ACCESS FOUNDATION

===
=========
11. USERS TABLE
===
=========

TABLE

users

PURPOSE

Stores every registered user.

COLUMNS

id UUID PRIMARY KEY

email VARCHAR(320) UNIQUE NOT NULL

password_hash TEXT NULL
(NULL when using OAuth only)

first_name VARCHAR(100)

last_name VARCHAR(100)

display_name VARCHAR(150)

avatar_url TEXT

language VARCHAR(10)

timezone VARCHAR(50)

status VARCHAR(20)

email_verified BOOLEAN

last_login_at TIMESTAMPTZ

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

==
12. USER STATUS
==

ALLOWED VALUES

ACTIVE

PENDING

SUSPENDED

DELETED

DEFAULT

PENDING

==
13. USER CONSTRAINTS
==

RULES

email

UNIQUE

NOT NULL

----

password_hash

Required only for password login.

----

email_verified

Default

FALSE

==
14. ORGANIZATIONS TABLE
==

TABLE

organizations

PURPOSE

Stores companies and teams.

COLUMNS

id UUID PRIMARY KEY

name VARCHAR(255)

slug VARCHAR(120)

country VARCHAR(2)

vat_number VARCHAR(50)

website TEXT

logo_url TEXT

created_at

updated_at

deleted_at

==
15. ORGANIZATION RULES
==

ONE ORGANIZATION

↓

Many Users

ONE ORGANIZATION

↓

Many Projects

==
16. ORGANIZATION MEMBERS TABLE

==
=========

TABLE

organization_members

PURPOSE

Connects users to organizations.

FIELDS

id UUID

organization_id UUID

user_id UUID

role VARCHAR(30)

joined_at TIMESTAMPTZ

==
=========
17. ORGANIZATION ROLES
==
=========

VALUES

OWNER

ADMIN

ARCHITECT

EDITOR

VIEWER

DEFAULT

VIEWER

==
=========
18. USER SESSIONS TABLE
==
=========

TABLE

user_sessions

PURPOSE

Tracks active logins.

FIELDS

id UUID

user_id UUID

refresh_token_hash TEXT

ip_address TEXT

device_name TEXT

user_agent TEXT

expires_at TIMESTAMPTZ

created_at TIMESTAMPTZ

revoked_at TIMESTAMPTZ

==
=========
19. USER PREFERENCES TABLE
==
=========

TABLE

user_preferences

PURPOSE

Stores user settings.

FIELDS

id UUID

user_id UUID

theme

language

measurement_unit

currency

default_project_template

autosave_interval

grid_size

snap_enabled

created_at

updated_at

==
20. MEASUREMENT SETTINGS
==

SUPPORTED

Metric

Imperial

DEFAULT

Metric

==
=========
21. CURRENCY SETTINGS
==
=========

SUPPORTED

EUR

RON

USD

GBP

HUF

DEFAULT

EUR

==
=========
22. AUTHENTICATION PROVIDERS
==
=========

SUPPORTED

Password

Google

Microsoft

Apple

GitHub

Stored in

user_auth_providers

===
=========

23. USER AUTH PROVIDERS TABLE

===
=========

FIELDS

id UUID

user_id UUID

provider

provider_user_id

created_at

===
=========

24. USER INDEXES

===
=========

INDEXES

users(email)

users(status)

organization_members(user_id)

organization_members(organization_id)

user_sessions(user_id)

===
=========

25. IDENTITY FOUNDATION COMPLETE

===
=========

NEXT SECTION

Projects

Project ownership

Project metadata

Versioning

===
END OF DATABASE SPECIFICATION PART 2
===

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 3

STATUS:
PROJECT FOUNDATION

===
26. PROJECTS TABLE
===

TABLE

projects

PURPOSE

Stores every architectural project.

FIELDS

id UUID PRIMARY KEY

organization_id UUID NULL

owner_id UUID NOT NULL

name VARCHAR(255)

description TEXT

status VARCHAR(30)

visibility VARCHAR(30)

project_type VARCHAR(30)

country_code CHAR(2)

city VARCHAR(120)

site_area DECIMAL(10,2)

building_area DECIMAL(10,2)

total_floor_area DECIMAL(10,2)

current_version_id UUID NULL

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

==
27. PROJECT STATUS
==

ALLOWED VALUES

DRAFT

IN_PROGRESS

UNDER_REVIEW

APPROVED

ARCHIVED

DEFAULT

DRAFT

===
=========
28. PROJECT VISIBILITY
===
=========

VALUES

PRIVATE

ORGANIZATION

PUBLIC_LINK

PUBLIC_READONLY

DEFAULT

PRIVATE

===
=========
29. PROJECT TYPE
===
=========

VALUES

HOUSE

APARTMENT

VILLA

OFFICE

COMMERCIAL

CUSTOM

==
=========
30. PROJECT OWNERSHIP RULES
==
=========

ONE USER

↓

Many Projects

ONE PROJECT

↓

One Owner

Owner may transfer ownership.

==
=========
31. PROJECT SETTINGS TABLE
==
=========

TABLE

project_settings

PURPOSE

Stores project-specific configuration.

FIELDS

id UUID

project_id UUID

measurement_unit

currency

grid_size

snap_to_grid BOOLEAN

auto_save BOOLEAN

auto_validation BOOLEAN

show_dimensions BOOLEAN

show_room_labels BOOLEAN

created_at

updated_at

==
=========
32. PROJECT METADATA TABLE
==
=========

TABLE

project_metadata

PURPOSE

Stores searchable metadata.

FIELDS

id UUID

project_id UUID

tags JSONB

keywords JSONB

notes TEXT

favorite BOOLEAN

last_opened_at TIMESTAMPTZ

==

33. PROJECT TAG EXAMPLES

==

EXAMPLES

modern

minimal

budget

family

garage

energy_a

wood

brick

smart_home

==

34. PROJECT FAVORITES

==

favorite BOOLEAN

DEFAULT

FALSE

Used for

Dashboard

Quick access

==
35. PROJECT VERSION TABLE
==

TABLE

project_versions

PURPOSE

Stores immutable project snapshots.

FIELDS

id UUID

project_id UUID

version_number INTEGER

version_name VARCHAR(100)

description TEXT

created_by UUID

snapshot JSONB

is_auto_save BOOLEAN

created_at TIMESTAMPTZ

==
36. VERSION RULES
==

Every manual save

↓

Creates new version

Every AI generation

↓

Creates new version

Auto-save

↓

Optional version

==
37. CURRENT VERSION POINTER
==

projects.current_version_id

Always points to

Latest active version

Allows fast loading

==
38. PROJECT PERMISSIONS TABLE
==

TABLE

project_permissions

FIELDS

id UUID

project_id UUID

user_id UUID

role

granted_by

created_at

===
=========
39. PROJECT PERMISSION ROLES
===
=========

VALUES

OWNER

EDITOR

COMMENTER

VIEWER

===
=========
40. PROJECT FOUNDATION COMPLETE
===
=========

NEXT SECTION

Room system

Geometry

Walls

Doors

Windows

==
=========
END OF DATABASE SPECIFICATION PART 3
==
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 4

STATUS:
ROOM SYSTEM FOUNDATION

==
=========
41. ROOMS TABLE
==
=========

TABLE

rooms

PURPOSE

Stores every room belonging to a project version.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID NOT NULL

parent_room_id UUID NULL

room_type VARCHAR(50)

room_name VARCHAR(150)

floor_number INTEGER

x DECIMAL(10,2)

y DECIMAL(10,2)

width DECIMAL(10,2)

height DECIMAL(10,2)

rotation DECIMAL(6,2)

ceiling_height DECIMAL(5,2)

area DECIMAL(10,2)

perimeter DECIMAL(10,2)

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

==
=========
42. ROOM TYPE VALUES
==
=========

SUPPORTED TYPES

BEDROOM

MASTER_BEDROOM

LIVING_ROOM

DINING_ROOM

KITCHEN

PANTRY

BATHROOM

WC

LAUNDRY

OFFICE

GARAGE

HALLWAY

STAIRS

STORAGE

DRESSING

TERRACE

BALCONY

UTILITY

CUSTOM

==
=========
43. ROOM POSITION SYSTEM
==
=========

COORDINATE SYSTEM

Origin

(0,0)

X

Left → Right

Y

Top → Bottom

UNIT

Centimeters (internal)

===
=========
44. ROOM DIMENSION RULES
===
=========

RULES

width > 0

height > 0

area > 0

Maximum values configurable.

===
=========
45. ROOM AREA CALCULATION
===
=========

FORMULA

area = width × height

Stored as cache.

Recalculated automatically after resize.

===
=========
46. ROOM PERIMETER
===
=========

FORMULA

perimeter =

2 × (width + height)

Auto-calculated.

===
=========
47. ROOM HIERARCHY
===
=========

parent_room_id

Allows

- nested areas
- split rooms
- zones

Normally NULL.

===
=========
48. ROOM ATTRIBUTES TABLE
===
=========

TABLE

room_attributes

FIELDS

id UUID

room_id UUID

heated BOOLEAN

air_conditioned BOOLEAN

natural_light BOOLEAN

requires_plumbing BOOLEAN

requires_ventilation BOOLEAN

fire_escape BOOLEAN

created_at

updated_at

===
=========
49. ROOM FINISHES TABLE
===
=========

TABLE

room_finishes

FIELDS

id UUID

room_id UUID

floor_material

wall_material

ceiling_material

paint_color

floor_color

created_at

updated_at

===
=========
50. ROOM USAGE STATISTICS
===
=========

TABLE

room_statistics

FIELDS

id UUID

room_id UUID

estimated_cost

estimated_build_time

energy_score

comfort_score

optimization_score

created_at

updated_at

===
=========
51. ROOM CONSTRAINTS TABLE
===
=========

TABLE

room_constraints

FIELDS

id UUID

room_id UUID

minimum_width

minimum_height

maximum_width

maximum_height

minimum_area

maximum_area

created_at

updated_at

==
=========
52. ROOM TAGS TABLE
==
=========

TABLE

room_tags

FIELDS

id UUID

room_id UUID

tag

EXAMPLES

private

public

wet_area

noise_sensitive

high_traffic

guest

==
=========

53. ROOM INDEXES
==
=========

INDEXES

rooms(project_version_id)

rooms(room_type)

rooms(floor_number)

room_tags(room_id)

room_attributes(room_id)

==
=========
54. ROOM RELATIONSHIPS
==
=========

ONE PROJECT VERSION

↓

Many Rooms

ONE ROOM

↓

One Statistics Row

ONE ROOM

↓

One Attributes Row

ONE ROOM

↓

Many Tags

==
=========
55. ROOM FOUNDATION COMPLETE
==
=========

NEXT SECTION

Walls

Corners

Geometry

Connections

==
=========
END OF DATABASE SPECIFICATION PART 4
==
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 5

STATUS:
WALL SYSTEM FOUNDATION

==
=========
56. WALLS TABLE
==
=========

TABLE

walls

PURPOSE

Stores every wall belonging to a room or building perimeter.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID NOT NULL

room_id UUID NULL

wall_type VARCHAR(50)

start_x DECIMAL(10,2)

start_y DECIMAL(10,2)

end_x DECIMAL(10,2)

end_y DECIMAL(10,2)

length DECIMAL(10,2)

thickness DECIMAL(6,2)

height DECIMAL(6,2)

rotation DECIMAL(6,2)

load_bearing BOOLEAN

external_wall BOOLEAN

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

===

57. WALL TYPES

===

SUPPORTED

EXTERIOR

INTERIOR

PARTITION

LOAD_BEARING

RETAINING

GARAGE

CUSTOM

==
=========
58. WALL GEOMETRY
==
=========

START POINT

(start_x, start_y)

END POINT

(end_x, end_y)

UNIT

Centimeters

==
=========
59. WALL LENGTH
==
=========

FORMULA

length =
$\sqrt{((end\_x-start\_x)^2+(end\_y-start\_y)^2)}$

Calculated automatically.

==
=========
60. WALL THICKNESS
==
=========

DEFAULT VALUES

Interior

10 cm

Structural

25 cm

Exterior

30–45 cm

Configurable.

==
=========
61. WALL MATERIALS TABLE
==
=========

TABLE

wall_materials

FIELDS

id UUID

wall_id UUID

material_type

density

thermal_conductivity

fire_rating

sound_rating

created_at

updated_at

==
=========
62. WALL CONNECTIONS TABLE
==
=========

TABLE

wall_connections

PURPOSE

Stores wall-to-wall joins.

FIELDS

id UUID

wall_a_id UUID

wall_b_id UUID

connection_type

created_at

==
=========
63. CONNECTION TYPES

==
=========

VALUES

END_TO_END

T_JUNCTION

CROSS

CORNER

CUSTOM

==
=========
64. WALL OPENINGS
==
=========

A wall may contain

↓

Doors

Windows

Future utilities

Relationship

ONE WALL

↓

Many Openings

==
=========
65. WALL ATTRIBUTES TABLE
==
=========

FIELDS

id UUID

wall_id UUID

is_insulated BOOLEAN

has_electrical BOOLEAN

has_plumbing BOOLEAN

has_heating BOOLEAN

requires_fire_protection BOOLEAN

created_at

updated_at

==
66. WALL COST TABLE
==

TABLE

wall_costs

FIELDS

id UUID

wall_id UUID

estimated_material_cost

estimated_labor_cost

estimated_total_cost

created_at

updated_at

==
=========
67. WALL VALIDATION RULES
==
=========

RULES

✓ length > 0

✓ thickness > 0

✓ valid coordinates

✓ connected if required

✓ no zero-length walls

==
=========
68. WALL INDEXES
==
=========

INDEXES

walls(project_version_id)

walls(room_id)

wall_connections(wall_a_id)

wall_connections(wall_b_id)

==
=========
69. WALL RELATIONSHIPS
==
=========

ONE ROOM

↓

Many Walls

ONE WALL

↓

Many Doors

ONE WALL

↓

Many Windows

ONE WALL

↓

One Material Record

==
=========
70. WALL FOUNDATION COMPLETE
==
=========

NEXT SECTION

Doors

Windows

Openings

Hardware

==
=========
END OF DATABASE SPECIFICATION PART 5
==
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 6

STATUS:
DOOR SYSTEM FOUNDATION

==
=========
71. DOORS TABLE
==
=========

TABLE

doors

PURPOSE

Stores every door in a project version.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID NOT NULL

wall_id UUID NOT NULL

room_from_id UUID NULL

room_to_id UUID NULL

door_type VARCHAR(50)

position_from_wall_start DECIMAL(10,2)

width DECIMAL(6,2)

height DECIMAL(6,2)

frame_thickness DECIMAL(6,2)

opening_direction VARCHAR(20)

hinge_side VARCHAR(10)

opening_angle DECIMAL(5,2)

is_fire_rated BOOLEAN

is_accessible BOOLEAN

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

===
=========
72. DOOR TYPES
===
=========

SUPPORTED

HINGED

DOUBLE_HINGED

SLIDING

POCKET

FOLDING

PIVOT

GARAGE

GLASS

CUSTOM

===
=========
73. OPENING DIRECTION
===
=========

VALUES

INWARD

OUTWARD

LEFT

RIGHT

BOTH

NONE

===
=========
74. HINGE SIDE
===
=========

VALUES

LEFT

RIGHT

CENTER

NONE

===
=========
75. STANDARD DOOR DIMENSIONS
===
=========

COMMON WIDTHS

60 cm

70 cm

80 cm

90 cm

100 cm

120 cm

COMMON HEIGHTS

200 cm

210 cm

220 cm

CUSTOM SIZES ALLOWED

===
=========
76. DOOR MATERIALS TABLE
===
=========

TABLE

door_materials

FIELDS

id UUID

door_id UUID

material_type

finish

color

manufacturer

model

estimated_cost

created_at

updated_at

===
=========
77. DOOR HARDWARE TABLE

===
=========

TABLE

door_hardware

FIELDS

id UUID

door_id UUID

handle_type

lock_type

closer_type

threshold_type

smart_lock BOOLEAN

created_at

updated_at

===
=========
78. DOOR VALIDATION RULES
===
=========

RULES

✓ width > 0

✓ height > 0

✓ positioned inside wall

✓ wall exists

✓ no overlap with another opening

===
=========
79. ROOM CONNECTION LOGIC
===
=========

A DOOR MAY CONNECT

Room A

↓

Room B

OR

Room

↓

Outside

(room_to_id = NULL)

===
=========
80. ACCESSIBILITY ATTRIBUTES
===
=========

FIELDS

minimum_clear_width

wheelchair_accessible

automatic_opening

low_threshold

Used for accessibility validation.

===
=========

81. FIRE SAFETY ATTRIBUTES
==

FIELDS

fire_rating

smoke_seal

emergency_exit

panic_bar

Optional.

==
82. DOOR INDEXES
==

INDEXES

doors(project_version_id)

doors(wall_id)

doors(room_from_id)

doors(room_to_id)

==
83. DOOR RELATIONSHIPS
==

ONE WALL

↓

Many Doors

ONE DOOR

↓

One Material Record

ONE DOOR

↓

One Hardware Record

===
=========
84. DOOR FOUNDATION COMPLETE
===
=========

NEXT SECTION

Windows

Glass

Natural Light

Ventilation

===
=========
END OF DATABASE SPECIFICATION PART 6
===
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 7

STATUS:
WINDOW SYSTEM FOUNDATION

===
=========
85. WINDOWS TABLE
===
=========

TABLE

windows

PURPOSE

Stores every window in a project version.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID NOT NULL

wall_id UUID NOT NULL

room_id UUID NULL

window_type VARCHAR(50)

position_from_wall_start DECIMAL(10,2)

sill_height DECIMAL(6,2)

width DECIMAL(6,2)

height DECIMAL(6,2)

frame_depth DECIMAL(6,2)

opening_type VARCHAR(30)

opening_direction VARCHAR(20)

glass_layers SMALLINT

is_operable BOOLEAN

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

===
=========
86. WINDOW TYPES
===
=========

SUPPORTED

FIXED

CASEMENT

AWNING

SLIDING

TILT

TILT_AND_TURN

BAY

SKYLIGHT

PANORAMIC

CUSTOM

===
=========
87. OPENING TYPES
===
=========

VALUES

NONE

SIDE_HINGED

TOP_HINGED

BOTTOM_HINGED

SLIDING

TILT

TILT_AND_TURN

==
=========
88. STANDARD WINDOW DIMENSIONS
==
=========

COMMON WIDTHS

60 cm

80 cm

100 cm

120 cm

150 cm

180 cm

COMMON HEIGHTS

60 cm

90 cm

120 cm

150 cm

180 cm

CUSTOM SIZES SUPPORTED

==
=========
89. WINDOW MATERIALS TABLE
==
=========

TABLE

window_materials

FIELDS

id UUID

window_id UUID

frame_material

glass_type

glass_coating

thermal_value

solar_factor

estimated_cost

created_at

updated_at

==
=========
90. WINDOW PERFORMANCE TABLE
==
=========

TABLE

window_performance

FIELDS

id UUID

window_id UUID

u_value

g_value

sound_rating

air_tightness

water_tightness

wind_resistance

created_at

updated_at

===

91. NATURAL LIGHT ATTRIBUTES

===

FIELDS

estimated_light_area

orientation

sun_exposure

daylight_factor

Calculated automatically where possible.

===

92. VENTILATION ATTRIBUTES

===

FIELDS

ventilation_area

max_open_angle

natural_ventilation

cross_ventilation_candidate

Used by optimization engine.

==
=========
93. WINDOW VALIDATION RULES
==
=========

RULES

✓ width > 0

✓ height > 0

✓ inside wall boundaries

✓ no overlap with doors

✓ no overlap with other windows

==
=========
94. WINDOW INDEXES
==
=========

INDEXES

windows(project_version_id)

windows(room_id)

windows(wall_id)

==
=========
95. WINDOW RELATIONSHIPS
==
=========

ONE WALL

↓

Many Windows

ONE WINDOW

↓

One Material Record

ONE WINDOW

↓

One Performance Record

===
=========
96. WINDOW FOUNDATION COMPLETE
===
=========

NEXT SECTION

Floors

Roofs

Stairs

Building Structure

===
=========
END OF DATABASE SPECIFICATION PART 7
===
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 8

STATUS:
FLOOR SYSTEM FOUNDATION

===
=========

97. FLOORS TABLE
==

TABLE

floors

PURPOSE

Stores every building floor (level) within a project.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID NOT NULL

floor_number INTEGER NOT NULL

floor_name VARCHAR(100)

elevation DECIMAL(8,2)

height DECIMAL(6,2)

usable_area DECIMAL(10,2)

gross_area DECIMAL(10,2)

description TEXT

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

==
98. FLOOR TYPES
==

SUPPORTED

BASEMENT

SEMI_BASEMENT

GROUND_FLOOR

MEZZANINE

FIRST_FLOOR

SECOND_FLOOR

ATTIC

ROOF_LEVEL

CUSTOM

===
=========
99. FLOOR NUMBERING
===
=========

EXAMPLES

-2

-1

0

1

2

3

...

GROUND FLOOR

Default

0

==
100. FLOOR GEOMETRY
==

FIELDS

usable_area

gross_area

height

AREA UNIT

Square meters

HEIGHT UNIT

Meters

==
101. FLOOR SETTINGS TABLE
==

TABLE

floor_settings

FIELDS

id UUID

floor_id UUID

default_wall_height

default_wall_thickness

default_ceiling_height

default_floor_material

default_ceiling_material

created_at

updated_at

==
102. FLOOR MATERIALS TABLE
==

TABLE

floor_materials

FIELDS

id UUID

floor_id UUID

material_type

finish

thermal_resistance

sound_insulation

estimated_cost

created_at

updated_at

==
103. FLOOR LOAD PARAMETERS

==
=========

TABLE

floor_loads

FIELDS

id UUID

floor_id UUID

live_load

dead_load

design_load

created_at

updated_at

Reserved for future engineering modules.

==
=========
104. FLOOR VALIDATION RULES
==
=========

RULES

✓ unique floor_number per project_version

✓ height > 0

✓ usable_area ≥ 0

✓ gross_area ≥ usable_area

==
=========
105. ROOM TO FLOOR RELATIONSHIP

==
=========

ONE FLOOR

↓

Many Rooms

----

rooms.floor_number

must match

floors.floor_number

==
=========
106. FLOOR INDEXES
==
=========

INDEXES

floors(project_version_id)

floors(floor_number)

floor_materials(floor_id)

==
=========
107. FLOOR RELATIONSHIPS
==
=========

ONE PROJECT VERSION

↓

Many Floors

ONE FLOOR

↓

Many Rooms

ONE FLOOR

↓

One Settings Record

===
=========
108. FLOOR FOUNDATION COMPLETE
===
=========

NEXT SECTION

Roof System

Roof Geometry

Roof Materials

===
=========
END OF DATABASE SPECIFICATION PART 8
===
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 9 + PART 10

STATUS:
ROOF SYSTEM + STAIR SYSTEM

===
=========
109. ROOFS TABLE
===
=========

TABLE

roofs

PURPOSE

Stores every roof belonging to a project version.

FIELDS

id UUID PRIMARY KEY

project_version_id UUID

roof_type

pitch

ridge_height

eave_height

surface_area

volume

material_id

created_at

updated_at

deleted_at

ROOF TYPES

GABLE

HIP

FLAT

SHED

GAMBREL

MANSARD

DOME

CUSTOM

==

110. ROOF MATERIALS

==

TABLE

roof_materials

FIELDS

id

name

manufacturer

weight_per_m2

thermal_resistance

expected_lifetime

maintenance_interval

estimated_cost_per_m2

SUPPORTED

Clay Tile

Concrete Tile

Metal

Standing Seam

Bitumen

PVC Membrane

EPDM

Slate

Wood Shingle

Green Roof

==
111. ROOF ATTRIBUTES
==

roof_attributes

snow_load

wind_load

solar_ready

green_roof

gutter_type

drainage_type

ventilation_type

==
112. ROOF VALIDATION
==

RULES

✓ pitch >=0°

✓ valid geometry

✓ material exists

✓ area >0

✓ connected to building

==

113. STAIRS TABLE
==

TABLE

stairs

PURPOSE

Stores all staircases.

FIELDS

id UUID

project_version_id

floor_from

floor_to

stair_type

width

height

total_length

step_count

landing_count

created_at

updated_at

==
114. STAIR TYPES
==

STRAIGHT

L_SHAPED

U_SHAPED

SPIRAL

CURVED

FLOATING

CUSTOM

===
=========
115. STAIR STEPS
===
=========

TABLE

stair_steps

FIELDS

id

stair_id

step_number

rise

run

width

material

===
=========
116. STAIR VALIDATION
===
=========

RULES

✓ rise >0

✓ run >0

✓ width >0

✓ minimum head clearance

✓ connects two valid floors

==
117. STAIR MATERIALS
==

SUPPORTED

Concrete

Steel

Wood

Glass

Stone

Composite

==
118. HANDRAILS
==

TABLE

handrails

FIELDS

id

stair_id

left_side

right_side

height

material

===
=========
119. STAIR INDEXES
===
=========

stairs(project_version_id)

stairs(floor_from)

stairs(floor_to)

===
=========
120. ROOF + STAIR FOUNDATION COMPLETE
===
=========

NEXT

Materials

Furniture

Building Components

===
=========

121. MATERIALS TABLE
===
=========

TABLE

materials

PURPOSE

Central material catalog.

FIELDS

id UUID

category

name

manufacturer

density

thermal_conductivity

fire_rating

sound_rating

carbon_footprint

unit

unit_cost

currency

created_at

updated_at

==
=========
122. MATERIAL CATEGORIES
==
=========

Concrete

Brick

Wood

Steel

Glass

Insulation

Roofing

Flooring

Paint

Tile

Stone

Composite

Custom

==
123. MATERIAL PRICE HISTORY
==

TABLE

material_prices

FIELDS

id

material_id

price

currency

valid_from

valid_to

supplier

==
124. MATERIAL SUPPLIERS
==

TABLE

material_suppliers

FIELDS

id

material_id

supplier_name

country

website

contact_email

rating

==

125. BUILDING COMPONENTS

==

TABLE

building_components

FIELDS

id

project_version_id

component_type

reference_id

material_id

quantity

unit

estimated_cost

==

126. COMPONENT TYPES

==

Wall

Door

Window

Roof

Floor

Stair

Column

Beam

Foundation

==
=========
127. COST SUMMARY CACHE
==
=========

TABLE

cost_summary

FIELDS

project_version_id

material_cost

labor_cost

equipment_cost

transport_cost

total_cost

last_calculated

==
=========
128. MATERIAL INDEXES
==
=========

materials(category)

materials(name)

material_prices(material_id)

building_components(project_version_id)

===
129. RELATIONSHIPS
===

ONE MATERIAL

↓

Many Prices

Many Components

ONE PROJECT

↓

Many Components

===
130. MATERIAL FOUNDATION COMPLETE
===

NEXT

Furniture

Electrical

Plumbing

HVAC

==
=========
END OF DATABASE SPECIFICATION PARTS 9 + 10
==
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 11 + PART 12

STATUS:
FURNITURE + MEP (ELECTRICAL, PLUMBING, HVAC)

==
=========
131. FURNITURE TABLE
==
=========

TABLE

furniture

FIELDS

id UUID PRIMARY KEY

project_version_id

room_id

category

name

manufacturer

model

x

y

rotation

width

depth

height

material_id

estimated_cost

created_at

updated_at

==
132. FURNITURE CATEGORIES
==

Bed

Sofa

Table

Chair

Wardrobe

Kitchen Cabinet

Bathroom Cabinet

Desk

TV Unit

Shelf

Appliance

Decoration

Custom

==
133. FURNITURE VALIDATION
==

RULES

✓ Inside room boundaries

✓ No collision (optional)

✓ Valid dimensions

✓ Rotation 0–360°

===
134. ELECTRICAL POINTS
===

TABLE

electrical_points

FIELDS

id

project_version_id

room_id

type

x

y

power_rating

voltage

created_at

updated_at

TYPES

Socket

Switch

Light

Ceiling Light

Wall Light

EV Charger

Network Port

==
135. ELECTRICAL CIRCUITS
==

TABLE

electrical_circuits

FIELDS

id

project_version_id

circuit_name

breaker_size

voltage

phase

max_load

==
136. PLUMBING FIXTURES
==

TABLE

plumbing_fixtures

FIELDS

id

project_version_id

room_id

fixture_type

water_supply

drain_size

x

y

==
137. FIXTURE TYPES
==

Sink

Toilet

Shower

Bathtub

Washing Machine

Dishwasher

Water Heater

Floor Drain

Outdoor Faucet

==
138. HVAC UNITS
==

TABLE

hvac_units

FIELDS

id

project_version_id

room_id

system_type

capacity_kw

airflow

energy_class

x

y

==
=========
139. HVAC TYPES
==
=========

Split AC

Multi Split

Heat Pump

Radiator

Underfloor Heating

Ventilation Unit

Fan Coil

Air Handling Unit

==
140. MEP VALIDATION
==

RULES

✓ Fixture belongs to room

✓ Electrical load valid

✓ HVAC capacity >0

✓ Plumbing connected

==

141. LIGHTING CALCULATIONS
==

TABLE

lighting_analysis

FIELDS

id

room_id

required_lux

installed_lux

lighting_power

efficiency

==
142. ENERGY CONSUMPTION
==

TABLE

energy_consumption

FIELDS

id

project_version_id

heating

cooling

lighting

hot_water

total_estimated

==
143. WATER CONSUMPTION
==

TABLE

water_consumption

FIELDS

id

project_version_id

daily_estimate

monthly_estimate

annual_estimate

==
144. ROOM SERVICES SUMMARY
==

TABLE

room_services

FIELDS

id

room_id

electrical_points

water_points

hvac_units

lighting_points

==
=========
145. SMART HOME DEVICES
==
=========

TABLE

smart_devices

FIELDS

id

project_version_id

room_id

device_type

manufacturer

protocol

==
=========

SUPPORTED

WiFi

Zigbee

Z-Wave

Matter

Bluetooth

==
=========
146. MEP INDEXES
==
=========

electrical_points(room_id)

plumbing_fixtures(room_id)

hvac_units(room_id)

smart_devices(room_id)

==
=========
147. RELATIONSHIPS
==
=========

ONE ROOM

↓

Many Furniture Items

Many Electrical Points

Many Plumbing Fixtures

Many HVAC Units

Many Smart Devices

==
=========
148. MEP FOUNDATION COMPLETE
==
=========

NEXT

AI History

Audit Logs

Notifications

Exports

==
=========
END OF DATABASE SPECIFICATION PARTS 11 + 12
==
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 13 + PART 14

STATUS:
AI SYSTEM + EXPORT + AUDIT + NOTIFICATIONS

==
=========
149. AI REQUESTS TABLE
==
=========

TABLE

ai_requests

FIELDS

id UUID PRIMARY KEY

project_id

project_version_id

user_id

ai_provider

ai_model

prompt

input_tokens

created_at

status

==
150. AI RESPONSES TABLE
==

TABLE

ai_responses

FIELDS

id UUID

request_id

response_json

output_tokens

processing_time_ms

validation_score

accepted

created_at

==
151. AI GENERATION HISTORY
==

Every AI generation stores

✓ Prompt

✓ Response

✓ JSON

✓ Processing time

✓ Model version

✓ User

===
152. AI VALIDATION RESULTS
===

TABLE

ai_validation

FIELDS

id

response_id

geometry_valid

rule_valid

cost_valid

errors JSONB

warnings JSONB

===
153. AI FEEDBACK
===

TABLE

ai_feedback

FIELDS

id

response_id

rating

comment

improved_version

created_at

==
=========
154. EXPORTS TABLE
==
=========

TABLE

exports

FIELDS

id

project_version_id

user_id

export_type

file_name

storage_path

file_size

status

created_at

==
=========
155. EXPORT TYPES
==
=========

PDF

JSON

PNG

SVG

DXF

IFC

OBJ

GLTF

ZIP

==
156. EXPORT HISTORY
==

Every export stores

✓ User

✓ Time

✓ Version

✓ Format

✓ File Size

✓ Success/Failure

==
157. AUDIT LOGS
==

TABLE

audit_logs

FIELDS

id

user_id

project_id

action

entity

entity_id

old_value JSONB

new_value JSONB

ip_address

created_at

==
=========
158. AUDIT ACTIONS
==
=========

LOGIN

LOGOUT

CREATE

UPDATE

DELETE

EXPORT

IMPORT

AI_GENERATE

AI_ACCEPT

AI_REJECT

==
=========

159. NOTIFICATIONS
==

TABLE

notifications

FIELDS

id

user_id

type

title

message

is_read

created_at

==
160. NOTIFICATION TYPES
==

Project Shared

AI Finished

Export Ready

Version Created

System Alert

Billing

Security

Update

==

161. USER ACTIVITY
===
=========

TABLE

user_activity

FIELDS

id

user_id

action

duration_seconds

device

browser

created_at

===
=========
162. API KEYS
===
=========

TABLE

api_keys

FIELDS

id

organization_id

name

hashed_key

permissions

expires_at

last_used

==
163. WEBHOOKS
==

TABLE

webhooks

FIELDS

id

organization_id

url

secret

events

active

==
164. SYSTEM EVENTS
==

TABLE

system_events

FIELDS

id

event_type

severity

message

metadata JSONB

created_at

===
165. INDEXES
===

ai_requests(project_id)

exports(project_version_id)

audit_logs(user_id)

notifications(user_id)

system_events(event_type)

===
166. DATABASE FOUNDATION STATUS
===

CORE DATABASE

✔ Users

✔ Organizations

✔ Projects

✔ Versions

✔ Rooms

✔ Walls

✔ Doors

✔ Windows

✔ Floors

✔ Roofs

✔ Materials

✔ Furniture

✔ MEP

✔ AI

✔ Audit

✔ Export

✔ Notifications

STATUS

DATABASE CORE: ~95% COMPLETE

NEXT

Advanced Indexing

Partitioning

Backup Strategy

Replication

Performance Optimization

===
=========
END OF DATABASE SPECIFICATION PARTS 13 + 14
===
=========

AI HOME DESIGNER
DATABASE SPECIFICATION
PART 15 + PART 16

STATUS:
PERFORMANCE + SECURITY + BACKUP + DATABASE OPERATIONS

===
=========
167. INDEXING STRATEGY
===
=========

INDEX TYPES

Primary Index

Unique Index

Composite Index

Partial Index

GIN (JSONB)

GiST (future GIS)

MOST USED INDEXES

users(email)

projects(owner_id)

projects(status)

project_versions(project_id)

rooms(project_version_id)

walls(project_version_id)

doors(wall_id)

windows(wall_id)

audit_logs(created_at)

==
168. SEARCH OPTIMIZATION
==

FULL TEXT SEARCH

Project Name

Description

Room Name

Tags

Notes

JSONB INDEX

AI Results

Metadata

Settings

==
=========
169. PARTITIONING
==
=========

LARGE TABLES

audit_logs

system_events

ai_requests

ai_responses

PARTITION

Monthly

Automatically created

==
=========
170. ARCHIVING
==
=========

OLDER THAN

2 years

↓

Archive Database

OLDER THAN

5 years

↓

Cold Storage

===
=========
171. BACKUP STRATEGY
===
=========

Incremental Backup

Every Hour

Full Backup

Daily

Encrypted Backup

Weekly

Disaster Backup

Monthly
(off-site)

===
=========
172. RESTORE STRATEGY

===
=========

SUPPORTED

Single Project Restore

Single User Restore

Database Restore

Point-in-Time Recovery

===
=========
173. DATABASE SECURITY
===
=========

ALL CONNECTIONS

TLS Required

Passwords

Argon2id Hash

Sensitive Fields

Encrypted

===
=========
174. ROW LEVEL SECURITY
===
=========

Users can access

Only their own projects

OR

Projects shared with them

OR

Organization projects

==
=========
175. TRANSACTION RULES
==
=========

Atomic Transactions

Required For

Project Save

AI Accept

Version Creation

Export Creation

==
=========
176. CONCURRENCY CONTROL
==
=========

METHOD

Optimistic Locking

FIELDS

version_number

updated_at

==
=========

177. PERFORMANCE TARGETS
==
=========

Login

<150 ms

Project Load

<500 ms

Project Save

<300 ms

AI Result Save

<250 ms

==
=========
178. DATABASE LIMITS
==
=========

MAX PROJECT SIZE

100,000 Objects

MAX ROOM COUNT

10,000

MAX WALL COUNT

100,000

MAX FILE SIZE

5 GB

==
179. MAINTENANCE TASKS
==

Nightly

VACUUM

ANALYZE

REINDEX (if needed)

Statistics Update

Backup Verification

==
180. DATABASE HEALTH
==

MONITOR

CPU

RAM

Connections

Replication Delay

Index Usage

Slow Queries

==
181. MIGRATION SYSTEM
==

Every Schema Change

↓

Migration File

↓

Version Number

↓

Rollback Script

==
182. DATABASE VERSIONING
==

FORMAT

v1.0.0

v1.1.0

v2.0.0

Every migration tracked.

==
183. DISASTER RECOVERY
==

RPO

≤ 15 minutes

RTO

≤ 60 minutes

==

184. DATABASE OPERATIONS COMPLETE
==
=========

STATUS

Database Layer

100% Complete

Production Ready

==
=========
185. NEXT DOCUMENT
==
=========

REST API SPECIFICATION

Contents:

• Authentication API
• Project API
• AI API
• Export API
• User API
• Admin API
• Webhooks
• Error Codes
• Rate Limits
• OpenAPI Specification

==
=========
END OF DATABASE SPECIFICATION
DATABASE DESIGN COMPLETE
==
=========